

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2023

Subject SA7 – Investment and Finance

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) You are an ALM (Asset Liability Management) Specialist, working in a commercial bank XYZ Ltd, mainly responsible for its

- liquidity risk management; and
- derivatives trading management

To assess / monitor liquidity risk, the Banking Regulator has prescribed a liquidity gap report with the following time buckets for expected future cash outflows and inflows arising from banking business:

Details	01-14D	15-28D	29D-3M	03-06M	06-12M	1-3Y	3-5Y	>5Y
A. Outflows								
B. Inflows								
Gap = B-A								

D=Days; M=Months; Y=Years

A derivatives trader in your team, during the course of a day, shorted 125 futures on ABC shares. He also took following options positions on the ABC shares whose current price is INR 100. The net aggregate premiums paid on options (i.e. total paid on longs – total received from shorts) was zero. All contracts will expire in one day. Each contract (lot size) has 2000 units. Total payoff, if ABC share price does not move, is INR 500,000.

No. of Calls	Strike Price (INR)	No. of Puts
Short 100	95	Short 50
Long 125	100	Long 125
Long 10	105	Short 110
Long 35 Calls	Net Exposure	Short 35 Puts

Your neighbour came to you seeking advice on Pre-Mature Withdrawal (PMW) of a cumulative (i.e. principal and total interest is paid at maturity) FD (Fixed Deposit) held in your bank. You learnt from your colleague in FD accounts department that for PMW of FD, the interest rate will be 0.50% less than the rate effective at the time of deposit for the duration the deposit was maintained with the bank.

FD Term structure of interest rates	0 to 1Y	1 to 2Y	2 to 3Y	(n-1) to nY
Interest rates in % p.a. At start of FD	X_1	X_2	X_3	X_n
Interest rates in % p.a. At PMW of FD	Y_1	Y_2	Y_3	Y_n

Suppose, at start, the FD term was 5 years and $X_5\%$ p.a. interest rate would be applicable for five years provided FD is hold with the bank until maturity. If PMW is done during year two (or three) then $X_2\% - 0.50\%$ (or $X_3\% - 0.50\%$) p.a. interest rate would be applicable for the duration the FD is held with the bank.

- List typical assets and liabilities found in balance sheet of a commercial bank. (3)
- In the context of banking business, define following mentioned risks in one or two lines:

- Credit risk
- Liquidity risk
- Margin Risk
- Re-pricing risk (2)

iii) Distinguish between the retail and wholesale banking activities / services provided to corporate customers. (5)

iv) Discuss in detail the issues or challenges:

- in liquidity risk management of banks; and
- in preparing (or using) the liquidity gap report.

Your answer should consider cash flows related to balance sheet items as well as off-balance sheet items related to future events / actions. (20)

v) Outline the structural balance sheet ratios used to assess / measure liquidity risk in banks. (5)

vi) For the derivatives traders' FnO (Futures and Options) positions on ABC shares, answer the following:

- determine average price at which the trader shorted 125 futures;
- calculate the total payoffs if share price moves +/- 5% at expiry; and
- comment on the reasonableness of two payoffs calculated above (5)

vii) Discuss on the PMW feature of FDs of the bank. Your answer should include:

- merits / demerits of the PMW feature; and
- suggest suitable changes in the PMW feature to mitigate market risk. (6)

viii) Instead of PMW of FD, which alternative facility available in your bank would you suggest to your neighbour on existing FD? What other ways / products would you suggest your neighbour to consider before making any new FDs in future? (4)

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Q. 2) ABC is a life insurance company operating in the country Actuarial. Based on the current global economic scenario, inflation is a major concern for most economies. As a life insurance company, the impact of inflation on the ABC's portfolio can be significant as it can erode the real value of the company's assets over time, making it challenging to meet the company's long-term obligations to policyholders. Additionally, as Environment, Social & Governance (ESG) investing becomes increasingly important, it is also important to consider the impact of inflation on sustainable investing strategies.

As an Investment Actuary leading the investment team, it is essential to design an investment strategy that can effectively mitigate the effects of inflation on the company's assets, while also aligning with the company's ESG goals and achieving its investment objectives.

- i) What are some of the key investment instruments and techniques that can be used to manage the impact of inflation on a life insurance company's investment portfolio, and what are their relative strengths and weaknesses? (10)
 - ii) How can an investment strategy that incorporates ESG considerations help a life insurance company mitigate the impact of inflation on its portfolio, and what are some of the key ESG factors that should be considered? (8)
 - iii) How can a life insurance company balance the need to effectively manage inflation risk with the need to achieve its investment objectives and meet the long-term needs of its policyholders? (12)
 - iv) How can you ensure that a life insurance company's investment strategy remains effective in managing inflation risk and aligning with its ESG goals over the long term, and what are some of the key monitoring and evaluation metrics that should be used? (10)
 - v) What are the key risks associated with inflation for a life insurance company's investment portfolio, and how can they impact the company's ability to meet its long-term obligations to policyholders? (10)
- [50]**
